

Lecture 1

Supervised Learning and Linear Models

Machine Learning I

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Outline

The Supervised Learning Problem

Linear Regression and Least Squares

The Geometry of Least Squares

Statistical Properties

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Generalization: Bias and Variance

Summary

The Supervised Learning Problem

The Data

We observe N input–output pairs

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N, \quad \mathbf{x}_i \in \mathbb{R}^p, \quad y_i \in \mathcal{Y},$$

drawn independently from an unknown joint distribution $P(\mathbf{x}, y)$.

- $\mathbf{x}_i$ is the **feature vector** (covariates, inputs, predictors).
- y_i is the **target** (label, response, output).
- $\mathcal{Y} = \mathbb{R}$: **regression**. $\mathcal{Y} = \{1, \dots, K\}$: **classification**.

Goal

Find a function $f: \mathbb{R}^p \rightarrow \mathcal{Y}$ such that $f(\mathbf{x}) \approx y$ for pairs we have not seen.

The last clause is the whole subject. Fitting the N points we have is easy; the difficulty is predicting the $(N+1)$ -th.

Two Basic Problems

Regression $y \in \mathbb{R}$

$$y = f(\mathbf{x}) + \varepsilon$$

The response is a noisy function of the input.

Example: predict a household's expenditure from its characteristics.

Classification $y \in \{1, \dots, K\}$

$$P(y \mid \mathbf{x})$$

The response is a category; we model its conditional distribution.

Example: is this e-mail spam? ($K = 2$)

Both are instances of the same recipe. This lecture treats regression; Lecture 2 turns to classification.

The Hypothesis Class

We cannot search over *all* functions $\mathbb{R}^p \rightarrow \mathcal{Y}$. We fix a **hypothesis class** $\mathcal{H}$ and search inside it.

$$\mathcal{H}_{\text{lin}} = \{ f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b : \mathbf{w} \in \mathbb{R}^p, b \in \mathbb{R} \}$$

- A choice of $\mathcal{H}$ is a choice of what patterns we are willing to represent.
- Too small: no $f \in \mathcal{H}$ fits the data well (**underfitting**).
- Too large: many $f \in \mathcal{H}$ fit the data, most of them badly elsewhere (**overfitting**).

Linear models are the smallest interesting $\mathcal{H}$. Nearly every method in this course is either a linear model in disguise or a controlled way of enlarging $\mathcal{H}_{\text{lin}}$.

The Loss Function

A **loss** $L(y, \hat{y}) \geq 0$ measures the cost of predicting $\hat{y}$ when the truth is y .

Name	$L(y, \hat{y})$	Used for
Squared	$(y - \hat{y})^2$	regression
Absolute	$ y - \hat{y} $	robust regression
0/1	$\mathbf{1}\{y \neq \hat{y}\}$	classification (the loss we <i>care</i> about)

The **empirical risk** of f on $\mathcal{D}$ is the average loss over the sample:

$$\hat{R}(f) = \frac{1}{N} \sum_{i=1}^N L(y_i, f(\mathbf{x}_i)).$$

The **true risk** is the expectation under the data-generating distribution:

$$R(f) = \mathbb{E}_{(\mathbf{x}, y) \sim P} [L(y, f(\mathbf{x}))].$$

The Recipe

Empirical Risk Minimization

$$\hat{f} = \arg \min_{f \in \mathcal{H}} \frac{1}{N} \sum_{i=1}^N L(y_i, f(\mathbf{x}_i))$$

Every supervised method in this course is an instance of this recipe, obtained by choosing three things:

1. a **hypothesis class** $\mathcal{H}$ (linear, trees, networks, ...)
2. a **loss** L (squared, hinge, log, exponential, ...)
3. an **optimizer** for the arg min (closed form, gradient descent, greedy, ...)

When we say “least squares”, “SVM”, “logistic regression”, “AdaBoost”, we are naming a triple.

Why the 0/1 Loss Cannot Be Optimized Directly

For a linear classifier $f(\mathbf{x}) = \text{sign}(\mathbf{w}^\top \mathbf{x} + b)$ and 0/1 loss, the empirical risk is

$$\hat{R}(\mathbf{w}, b) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{y_i \neq \text{sign}(\mathbf{w}^\top \mathbf{x}_i + b)\}.$$

This objective has no usable gradient

As a function of $(\mathbf{w}, b)$, $\hat{R}$ is **piecewise constant**: it takes values in $\{0, \frac{1}{N}, \frac{2}{N}, \dots, 1\}$ and changes only when some $\mathbf{w}^\top \mathbf{x}_i + b$ crosses zero.

- Its gradient is **0** almost everywhere and undefined on the boundaries.
- It is not convex: a local minimum tells us nothing about the global one.
- Minimizing it exactly is NP-hard in general.

This is why classification methods replace the 0/1 loss by a **convex surrogate** (hinge, log, exponential). Lecture 2 develops each one.

Linear Regression and Least Squares

The Linear Model

$$f(\mathbf{x}) = \beta_0 + \sum_{j=1}^p x_j \beta_j$$

The inputs x_j may be raw measurements or anything computed from them: transformations ($\log x$, x^2), basis expansions, dummy codes, interactions $x_1 x_2$. **Linear** refers to the parameters, not to the inputs.

Absorb the intercept by prepending a constant 1 to every input:

$$\mathbf{x}_i \leftarrow (1, x_{i1}, \dots, x_{ip})^\top \in \mathbb{R}^{p+1}, \quad \boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_p)^\top \in \mathbb{R}^{p+1}, \quad f(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}.$$

Stack the data:

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \vdots \\ \mathbf{x}_N^\top \end{pmatrix} \in \mathbb{R}^{N \times (p+1)}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix} \in \mathbb{R}^N, \quad \hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta}.$$

The Least Squares Objective

With squared loss, empirical risk minimization becomes

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^{p+1}} \text{RSS}(\beta), \quad \text{RSS}(\beta) = \sum_{i=1}^N (y_i - \mathbf{x}_i^\top \beta)^2.$$

In matrix form,

$$\text{RSS}(\beta) = (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) = \|\mathbf{y} - \mathbf{X}\beta\|^2.$$

The factor $\frac{1}{N}$ does not change the minimizer; we drop it here and reinstate it whenever a gradient step size needs to be scale-free.

Two Facts About Derivatives of Quadratic Forms

For $\mathbf{a} \in \mathbb{R}^d$ and symmetric $\mathbf{A} \in \mathbb{R}^{d \times d}$, with ∇_{β} the column vector of partial derivatives:

Linear term

$$\nabla_{\beta}(\mathbf{a}^{\top} \beta) = \mathbf{a}.$$

Proof. $\mathbf{a}^{\top} \beta = \sum_k a_k \beta_k$, so $\partial/\partial\beta_j$ picks out a_j . □

Quadratic term

$$\nabla_{\beta}(\beta^{\top} \mathbf{A} \beta) = 2\mathbf{A}\beta.$$

Proof. $\beta^{\top} \mathbf{A} \beta = \sum_{k,l} \beta_k A_{kl} \beta_l$. Differentiating with respect to β_j collects the terms with $k = j$ and the terms with $l = j$: $\sum_l A_{jl} \beta_l + \sum_k \beta_k A_{kj} = (\mathbf{A}\beta)_j + (\mathbf{A}^{\top} \beta)_j = 2(\mathbf{A}\beta)_j$ by symmetry. □

These two identities carry the whole course.

Deriving the Normal Equations

Expand, then differentiate

$$\begin{aligned}\text{RSS}(\beta) &= (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta) \\ &= \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}\beta - \beta^\top \mathbf{X}^\top \mathbf{y} + \beta^\top \mathbf{X}^\top \mathbf{X}\beta \\ &= \mathbf{y}^\top \mathbf{y} - 2(\mathbf{X}^\top \mathbf{y})^\top \beta + \beta^\top (\mathbf{X}^\top \mathbf{X})\beta \quad (\text{middle terms: equal scalars})\end{aligned}$$

$$\begin{aligned}\nabla_\beta \text{RSS}(\beta) &= -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\beta \quad (\mathbf{a} = \mathbf{X}^\top \mathbf{y}, \mathbf{A} = \mathbf{X}^\top \mathbf{X}) \\ &= -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\beta).\end{aligned}$$

Setting the gradient to zero gives the **normal equations**:

$$\boxed{\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{y}} \implies \hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad \text{when } \mathbf{X}^\top \mathbf{X} \text{ is invertible.}$$

Is It a Minimum?

A stationary point of a smooth function is a minimum if the Hessian is positive definite there.

The Hessian of RSS

$$\nabla_{\beta}^2 \text{RSS}(\beta) = \nabla_{\beta} [-2\mathbf{X}^{\top} \mathbf{y} + 2\mathbf{X}^{\top} \mathbf{X} \beta] = 2\mathbf{X}^{\top} \mathbf{X}.$$

For any $\mathbf{v} \in \mathbb{R}^{p+1}$,

$$\mathbf{v}^{\top} (\mathbf{X}^{\top} \mathbf{X}) \mathbf{v} = (\mathbf{X} \mathbf{v})^{\top} (\mathbf{X} \mathbf{v}) = \|\mathbf{X} \mathbf{v}\|^2 \geq 0,$$

so $\mathbf{X}^{\top} \mathbf{X}$ is positive semidefinite, and positive definite exactly when $\mathbf{X} \mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v} = \mathbf{0}$, i.e. when the columns of $\mathbf{X}$ are linearly independent.

RSS is convex. When $\text{rank } \mathbf{X} = p + 1$ it is strictly convex and $\hat{\beta}$ is the unique global minimizer. There are no other local minima to worry about.

Numerical Example

Five observations, one feature: $x = (1, 2, 3, 4, 5)$, $y = (2.2, 2.8, 4.5, 3.7, 5.5)$.

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{y} = \begin{pmatrix} \sum y_i \\ \sum x_i y_i \end{pmatrix} = \begin{pmatrix} 18.7 \\ 63.6 \end{pmatrix}.$$

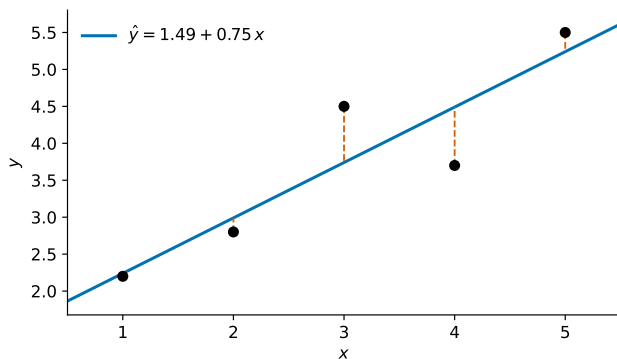
Solve the 2×2 system

$\det(\mathbf{X}^\top \mathbf{X}) = 5 \cdot 55 - 15^2 = 50$, so

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{50} \begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix}, \quad \hat{\beta} = \frac{1}{50} \begin{pmatrix} 55 \cdot 18.7 - 15 \cdot 63.6 \\ -15 \cdot 18.7 + 5 \cdot 63.6 \end{pmatrix} = \frac{1}{50} \begin{pmatrix} 74.5 \\ 37.5 \end{pmatrix} = \begin{pmatrix} 1.49 \\ 0.75 \end{pmatrix}.$$

Fitted line: $\hat{y} = 1.49 + 0.75x$.

Numerical Example: The Fit



Residuals $r_i = y_i - \hat{y}_i$: $(-0.04, -0.19, 0.76, -0.79, 0.26)$. They sum to zero — not by accident (next section).

Rank Deficiency

If the columns of $\mathbf{X}$ are linearly dependent (e.g. one dummy per category *and* an intercept):

- $\mathbf{X}^\top \mathbf{X}$ is singular; the normal equations have infinitely many solutions.
- $\hat{\boldsymbol{\beta}}$ is not identified, but $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ is — every solution gives the same fitted values, because they are all the same projection (next section).
- Numerically, near-dependence (multicollinearity) makes $(\mathbf{X}^\top \mathbf{X})^{-1}$ huge and $\hat{\boldsymbol{\beta}}$ unstable.

Never compute $(\mathbf{X}^\top \mathbf{X})^{-1}$ explicitly. Solve the linear system $\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{y}$ (Cholesky) or use the QR / SVD of $\mathbf{X}$. Lecture 3 (ridge) gives the principled fix for the singular case.

The Geometry of Least Squares

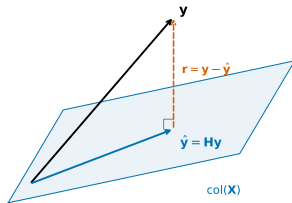
Least Squares Is a Projection

Write $\mathbf{X} = (\mathbf{x}^{(0)}, \mathbf{x}^{(1)}, \dots, \mathbf{x}^{(p)})$ by columns. Then

$$\mathbf{X}\boldsymbol{\beta} = \sum_{j=0}^p \beta_j \mathbf{x}^{(j)} \in \text{col}(\mathbf{X}),$$

the subspace of $\mathbb{R}^N$ spanned by the columns. Minimizing $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ over $\boldsymbol{\beta}$ means:

find the point of $\text{col}(\mathbf{X})$ closest to $\mathbf{y}$ — the **orthogonal projection** of $\mathbf{y}$ onto $\text{col}(\mathbf{X})$.



The Hat Matrix

Substituting $\hat{\beta}$ back:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\beta} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \mathbf{H}\mathbf{y}, \quad \boxed{\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top} \in \mathbb{R}^{N \times N}.$$

H is symmetric and idempotent

$$\begin{aligned}\mathbf{H}^\top &= (\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top)^\top = \mathbf{X}((\mathbf{X}^\top \mathbf{X})^{-1})^\top \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{H}, \\ \mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \underbrace{\mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}}_{=\mathbf{I}} \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{H}.\end{aligned}$$

The first uses $(\mathbf{ABC})^\top = \mathbf{C}^\top \mathbf{B}^\top \mathbf{A}^\top$ and that the inverse of a symmetric matrix is symmetric.

Symmetric + idempotent is exactly the definition of an orthogonal projection matrix.
Projecting twice changes nothing.

Residuals Are Orthogonal to the Fit

The residual vector is $\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$.

$$\mathbf{X}^\top \mathbf{r} = \mathbf{0}$$

$$\mathbf{X}^\top \mathbf{r} = \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{y} = \mathbf{0} \quad (\text{normal equations}).$$

Consequences:

- $\mathbf{r}$ is orthogonal to every column of $\mathbf{X}$, hence to all of $\text{col}(\mathbf{X})$, hence to $\hat{\mathbf{y}}$.
- With an intercept, the first column is $\mathbf{1}$, so $\mathbf{1}^\top \mathbf{r} = \sum_i r_i = 0$: residuals average to zero.
- Pythagoras in $\mathbb{R}^N$: $\|\mathbf{y}\|^2 = \|\hat{\mathbf{y}}\|^2 + \|\mathbf{r}\|^2$.

$\mathbf{I} - \mathbf{H}$ is also a projection — onto the orthogonal complement $\text{col}(\mathbf{X})^\perp$.

Degrees of Freedom

$$\text{tr } \mathbf{H} = p + 1$$

Using $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$,

$$\text{tr } \mathbf{H} = \text{tr}(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top) = \text{tr}((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}) = \text{tr}(\mathbf{I}_{p+1}) = p + 1.$$

- A projection matrix has eigenvalues in $\{0, 1\}$ (since $\mathbf{H}^2 = \mathbf{H} \Rightarrow \lambda^2 = \lambda$), so its trace counts the dimension of the subspace it projects onto.
- $\text{tr } \mathbf{H}$ is the number of parameters — the **degrees of freedom** of the fit.
- Correspondingly $\text{tr}(\mathbf{I} - \mathbf{H}) = N - p - 1$: the residual has $N - p - 1$ free directions. This is where the $N - p - 1$ in $\hat{\sigma}^2$ comes from.

Statistical Properties

The Gaussian Model

So far no probability. Now assume the data were generated by

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N),$$

with $\mathbf{X}$ fixed and $\boldsymbol{\beta}$, σ^2 unknown. The errors are independent, mean zero, common variance.

Three questions:

1. Is $\hat{\boldsymbol{\beta}}$ centred on $\boldsymbol{\beta}$? (bias)
2. How much does $\hat{\boldsymbol{\beta}}$ move from sample to sample? (variance)
3. Why squared loss and not something else? (likelihood)

Unbiasedness

$$\mathbb{E}[\hat{\beta}] = \beta$$

Substitute the model into the estimator:

$$\begin{aligned}\hat{\beta} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X} \beta + \varepsilon) \\ &= \underbrace{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}}_{=\mathbf{I}} \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \varepsilon.\end{aligned}$$

Taking expectations, with $\mathbf{X}$ fixed and $\mathbb{E}[\varepsilon] = \mathbf{0}$,

$$\mathbb{E}[\hat{\beta}] = \beta + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}[\varepsilon] = \beta.$$

The estimator equals the truth plus a linear transform of the noise. Everything below follows from that line.

Variance

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$$

Let $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ so that $\hat{\beta} - \beta = \mathbf{A}\varepsilon$. For a linear map of a random vector, $\text{Var}(\mathbf{A}\varepsilon) = \mathbf{A} \text{Var}(\varepsilon) \mathbf{A}^\top$. Hence

$$\begin{aligned} \text{Var}(\hat{\beta}) &= \mathbf{A} (\sigma^2 \mathbf{I}) \mathbf{A}^\top = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}. \end{aligned}$$

Since $\hat{\beta}$ is an affine function of Gaussian ε , it is Gaussian:

$$\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$$

The variance is driven by $(\mathbf{X}^\top \mathbf{X})^{-1}$. Nearly collinear columns make it large: that is multicollinearity, stated precisely.

Estimating σ^2

$$\mathbb{E} \|\mathbf{r}\|^2 = (N - p - 1)\sigma^2$$

$\mathbf{r} = (\mathbf{I} - \mathbf{H})\mathbf{y} = (\mathbf{I} - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$, because $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{X} - \mathbf{X} = \mathbf{0}$. Then, using $\mathbf{I} - \mathbf{H}$ symmetric idempotent and $\mathbb{E}[\boldsymbol{\varepsilon}^\top \mathbf{M} \boldsymbol{\varepsilon}] = \sigma^2 \text{tr} \mathbf{M}$ for fixed $\mathbf{M}$,

$$\mathbb{E} \|\mathbf{r}\|^2 = \mathbb{E}[\boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{H})^\top (\mathbf{I} - \mathbf{H}) \boldsymbol{\varepsilon}] = \mathbb{E}[\boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{H}) \boldsymbol{\varepsilon}] = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{H}) = \sigma^2(N - p - 1).$$

Therefore the **unbiased** estimator is

$$\hat{\sigma}^2 = \frac{1}{N - p - 1} \sum_{i=1}^N (y_i - \hat{y}_i)^2, \quad (N - p - 1) \hat{\sigma}^2 / \sigma^2 \sim \chi_{N-p-1}^2,$$

and $\hat{\boldsymbol{\beta}}$, $\hat{\sigma}^2$ are independent (they live in orthogonal subspaces, $\text{col } \mathbf{X}$ and $\text{col } \mathbf{X}^\perp$).

Where Does Squared Loss Come From?

Under the Gaussian model, $y_i \mid \mathbf{x}_i \sim \mathcal{N}(\mathbf{x}_i^\top \boldsymbol{\beta}, \sigma^2)$, so each observation has density

$$p(y_i \mid \mathbf{x}_i; \boldsymbol{\beta}, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{2\sigma^2}\right).$$

Maximum likelihood = least squares

By independence the likelihood is a product; take logs:

$$\begin{aligned}\ell(\boldsymbol{\beta}, \sigma^2) &= \sum_{i=1}^N \log p(y_i \mid \mathbf{x}_i) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 \\ &= -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \text{RSS}(\boldsymbol{\beta}).\end{aligned}$$

For any fixed $\sigma^2 > 0$, maximizing ℓ over $\boldsymbol{\beta}$ is minimizing $\text{RSS}(\boldsymbol{\beta})$.

Squared loss is not a convention. It is the negative log-likelihood of Gaussian noise. Absolute loss corresponds to Laplace noise; Lecture 2 derives log loss from Bernoulli in the same way.

The MLE of σ^2 Is Biased

Maximize ℓ over σ^2 at $\beta = \hat{\beta}$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{\text{RSS}(\hat{\beta})}{2\sigma^4} = 0 \implies \hat{\sigma}_{\text{MLE}}^2 = \frac{\text{RSS}(\hat{\beta})}{N}.$$

Compare with the unbiased $\hat{\sigma}^2 = \text{RSS}(\hat{\beta}) / (N - p - 1)$:

$$\mathbb{E}[\hat{\sigma}_{\text{MLE}}^2] = \frac{N - p - 1}{N} \sigma^2 < \sigma^2.$$

The MLE ignores that $p + 1$ degrees of freedom were spent fitting $\hat{\beta}$. The discrepancy is small for $N \gg p$ and the first hint of a theme: **training error underestimates true error**.

Gradient Descent

Why Not Always Use the Closed Form?

- Solving the normal equations costs $O(Np^2 + p^3)$: fine for p in the hundreds, not the millions.
- Most losses in this course (log, hinge, exponential) have *no* closed-form minimizer.
- Neural networks have none either.

We need a general-purpose optimizer. The simplest one uses only the gradient.

$$\theta^* = \arg \min_{\theta \in \mathbb{R}^d} L(\theta)$$

The Update Rule, From Taylor's Theorem

Which small step decreases L the most?

For a small displacement δ ,

$$L(\theta + \delta) \approx L(\theta) + \nabla L(\theta)^\top \delta.$$

Among all δ with fixed length $\|\delta\| = \eta$, the Cauchy–Schwarz inequality gives

$$\nabla L(\theta)^\top \delta \geq -\|\nabla L(\theta)\| \|\delta\|,$$

with equality iff δ points along $-\nabla L(\theta)$. The steepest descent direction is the negative gradient.

Gradient descent

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla L(\theta^{(t)}), \quad \eta > 0 \text{ the step size (learning rate).}$$

Plugging $\delta = -\eta \nabla L$ into the expansion: $L(\theta^{(t+1)}) \approx L(\theta^{(t)}) - \eta \|\nabla L\|^2$. Each step decreases the loss, if η is small enough for the approximation to hold.

Gradient Descent for Least Squares

With $L(\boldsymbol{\beta}) = \frac{1}{2N} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ (the $\frac{1}{2N}$ makes the step size scale-free),

$$\nabla L(\boldsymbol{\beta}) = -\frac{1}{N} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = -\frac{1}{N} \sum_{i=1}^N (y_i - \mathbf{x}_i^\top \boldsymbol{\beta}) \mathbf{x}_i.$$

Update

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \frac{\eta}{N} \sum_{i=1}^N (y_i - \mathbf{x}_i^\top \boldsymbol{\beta}^{(t)}) \mathbf{x}_i$$

Read it: every example pushes $\boldsymbol{\beta}$ in the direction of its own input $\mathbf{x}_i$, weighted by how badly it is currently predicted. Examples fit exactly contribute nothing.

How Large Can the Step Be?

Take the simplest case, $L(\theta) = \frac{1}{2}a\theta^2$ with $a > 0$, minimum at $\theta^* = 0$.

Exact dynamics

$\nabla L(\theta) = a\theta$, so

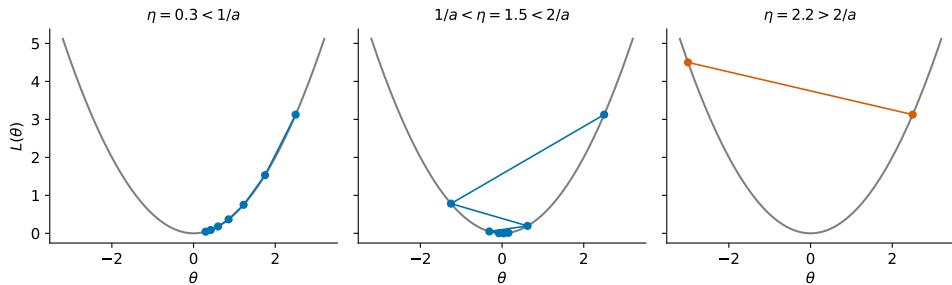
$$\theta^{(t+1)} = \theta^{(t)} - \eta a \theta^{(t)} = (1 - \eta a) \theta^{(t)} \implies \theta^{(t)} = (1 - \eta a)^t \theta^{(0)}.$$

The iterates converge iff $|1 - \eta a| < 1$, i.e. $0 < \eta < 2/a$.

- $\eta < 1/a$: monotone approach.
- $1/a < \eta < 2/a$: overshoots and oscillates, but still converges.
- $\eta > 2/a$: diverges geometrically.

For a general quadratic $\frac{1}{2}\theta^\top \mathbf{A}\theta$ the same argument runs along each eigenvector of $\mathbf{A}$: the binding constraint is $\eta < 2/\lambda_{\max}(\mathbf{A})$, and the slowest direction converges at rate $|1 - \eta\lambda_{\min}|$. The ratio $\lambda_{\max}/\lambda_{\min}$ — the condition number — governs speed.

Step Size in Pictures



Same quadratic, three step sizes. The trajectory on the right is what a too-large η looks like in practice: the loss goes *up*.

Stochastic Gradient Descent

The full gradient is an average over N terms. Replace it by one term chosen at random:

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \eta (y_{i_t} - \mathbf{x}_{i_t}^\top \boldsymbol{\beta}^{(t)}) \mathbf{x}_{i_t}, \quad i_t \sim \text{Uniform}\{1, \dots, N\}.$$

The stochastic gradient is unbiased

Let $g_i(\boldsymbol{\beta}) = -(y_i - \mathbf{x}_i^\top \boldsymbol{\beta}) \mathbf{x}_i$ be the gradient of the i -th term. Then

$$\mathbb{E}_{i_t} [g_{i_t}(\boldsymbol{\beta})] = \sum_{i=1}^N \frac{1}{N} g_i(\boldsymbol{\beta}) = \nabla L(\boldsymbol{\beta}).$$

So each step moves in the right direction *on average*, at $1/N$ of the cost. The price is variance: individual steps can point anywhere.

Mini-Batch SGD

Average over a random subset $\mathcal{B}_t \subset \{1, \dots, N\}$ of size B :

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - \frac{\eta}{B} \sum_{i \in \mathcal{B}_t} g_i(\boldsymbol{\beta}^{(t)}).$$

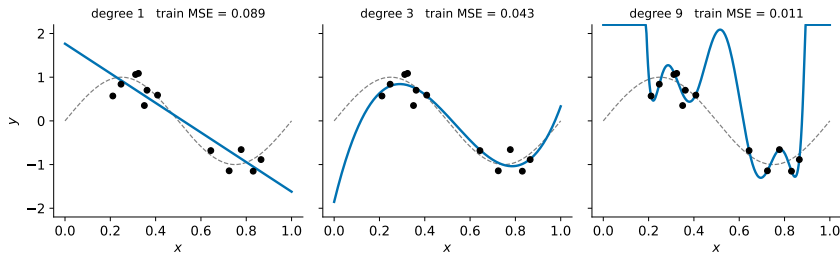
- Still unbiased, by the same argument.
- Variance of the step shrinks by a factor B (independent draws).
- $B = 1$ is SGD; $B = N$ is full gradient descent; $B \in [32, 256]$ is what everyone uses.

One pass through the data is an **epoch**. With a constant η , SGD does not converge to $\hat{\boldsymbol{\beta}}$; it hovers in a neighbourhood of size $\propto \eta$. Decaying η over epochs (e.g. $\eta_t \propto 1/t$) is what makes it converge.

Generalization: Bias and Variance

Training Error Is Not the Point

We minimize $\hat{R}(f)$, the error on $\mathcal{D}$. We care about $R(f)$, the error on new data.



Polynomial regression on the same 12 points. Degree 9 has the lowest training error and the worst predictions. The class $\mathcal{H}$ was too rich for the sample.

The Setup for Bias–Variance

Fix a test input $\mathbf{x}_0$. The truth is

$$y_0 = f(\mathbf{x}_0) + \varepsilon_0, \quad \mathbb{E}[\varepsilon_0] = 0, \quad \text{Var}(\varepsilon_0) = \sigma^2.$$

Our estimate $\hat{f}$ was trained on a random sample $\mathcal{D}$; it is itself random, with

$$\bar{f}(\mathbf{x}_0) = \mathbb{E}_{\mathcal{D}}[\hat{f}(\mathbf{x}_0)]$$

its average prediction over all training sets of size N . We study the expected test error at $\mathbf{x}_0$:

$$\mathbb{E}_{\mathcal{D}, \varepsilon_0} \left[(y_0 - \hat{f}(\mathbf{x}_0))^2 \right].$$

Two sources of randomness, assumed independent: the noise in y_0 , and the training sample.

The Decomposition

Write $f = f(\mathbf{x}_0)$, $\hat{f} = \hat{f}(\mathbf{x}_0)$, $\bar{f} = \bar{f}(\mathbf{x}_0)$.

Split off the noise, then split off the mean

$$\begin{aligned}\mathbb{E}[(y_0 - \hat{f})^2] &= \mathbb{E}[(f + \varepsilon_0 - \hat{f})^2] = \mathbb{E}[(f - \hat{f})^2] + \mathbb{E}[\varepsilon_0^2] + 2 \mathbb{E}[\varepsilon_0(f - \hat{f})] \\ &= \mathbb{E}[(f - \hat{f})^2] + \sigma^2 + 2 \underbrace{\mathbb{E}[\varepsilon_0] \mathbb{E}[f - \hat{f}]}_0\end{aligned}\quad (\varepsilon_0 \perp \mathcal{D})$$

$$\mathbb{E}[(f - \hat{f})^2] = \mathbb{E}[(f - \bar{f} + \bar{f} - \hat{f})^2] = (f - \bar{f})^2 + \mathbb{E}[(\bar{f} - \hat{f})^2] + 2(f - \bar{f}) \underbrace{\mathbb{E}[\bar{f} - \hat{f}]}_0.$$

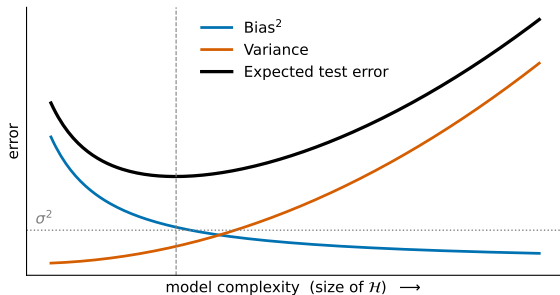
$$\mathbb{E}[(y_0 - \hat{f})^2] = \underbrace{(f - \bar{f})^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}[(\bar{f} - \hat{f})^2]}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Noise}}$$

Reading the Three Terms

Bias² How far the *average* model is from the truth. Zero if $f \in \mathcal{H}$ and the estimator is unbiased. Large when $\mathcal{H}$ cannot represent f .

Variance How much $\hat{f}$ jumps around across training sets. Large when $\mathcal{H}$ is rich relative to N : many members fit the sample, and which one we get depends on the sample.

Noise σ^2 . Irreducible. No method can beat it.



Bias–Variance for Least Squares

When $f(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}$ is truly linear and $\mathcal{D}$ is drawn from the Gaussian model:

- $\mathbb{E}[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta}$ (proved above), so $\bar{f}(\mathbf{x}_0) = \mathbf{x}_0^\top \boldsymbol{\beta} = f(\mathbf{x}_0)$: **bias is zero**.
- $\text{Var}(\hat{f}(\mathbf{x}_0)) = \text{Var}(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$.

Averaging the variance over the training inputs $\mathbf{x}_0 \in \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$:

$$\frac{1}{N} \sum_{i=1}^N \sigma^2 \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i = \frac{\sigma^2}{N} \text{tr}(\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top) = \frac{\sigma^2}{N} \text{tr} \mathbf{H} = \sigma^2 \frac{p+1}{N}.$$

Expected test error of least squares $\approx \sigma^2(1 + \frac{p+1}{N})$. Variance grows linearly in the number of parameters and shrinks with N . Every extra feature costs σ^2/N — unless it reduces bias by more.

The Trade-Off, and What To Do About It

Increasing the capacity of $\mathcal{H}$ (more features, higher-degree polynomials, deeper trees):

Bias² ↓ Variance ↑ Noise unchanged.

The sum has an interior minimum. Finding it is the practical content of the next lectures:

- **Lecture 3** — shrink $\hat{\beta}$ toward zero (ridge, lasso): accept a little bias to cut variance.
- **Lecture 4** — average many high-variance models (bagging): cut variance without adding bias.
- **Lecture 5** — add low-bias models one at a time (boosting): cut bias under control.

The decomposition is a statement about *expectations over training sets*. On one sample we never observe bias or variance separately — we observe only their sum, and only on held-out data.

Measuring Generalization: The Protocol

Since training error is optimistic, hold data out.

Split	Used for	Touch it how often?
Training	fitting $\hat{\beta}$	every iteration
Validation	choosing $\mathcal{H}$, λ , η , stopping time	every candidate
Test	reporting the final number	once

- Any decision made by looking at a split turns that split into training data.
- K -fold cross-validation reuses the training set as a rotating validation set; Lab 1 implements it.
- The test set is a one-shot instrument. Reuse it and its number stops meaning anything.

Summary

What We Proved Today

1. $\nabla_{\beta} \text{RSS} = -2\mathbf{X}^{\top}(\mathbf{y} - \mathbf{X}\beta)$ and $\nabla^2 \text{RSS} = 2\mathbf{X}^{\top}\mathbf{X} \succeq 0$: least squares is convex with unique minimizer $\hat{\beta} = (\mathbf{X}^{\top}\mathbf{X})^{-1}\mathbf{X}^{\top}\mathbf{y}$.
2. $\mathbf{H} = \mathbf{X}(\mathbf{X}^{\top}\mathbf{X})^{-1}\mathbf{X}^{\top}$ is symmetric, idempotent, $\text{tr} \mathbf{H} = p + 1$: the fit is a projection onto $\text{col}(\mathbf{X})$ and residuals are orthogonal to it.
3. Under Gaussian noise, $\hat{\beta} \sim \mathcal{N}(\beta, \sigma^2(\mathbf{X}^{\top}\mathbf{X})^{-1})$, and least squares is maximum likelihood.
4. Gradient descent is steepest descent from Taylor's theorem; on a quadratic it converges iff $\eta < 2/\lambda_{\max}$.
5. $\mathbb{E}[(y_0 - \hat{f})^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$, and for least squares the variance term is $\sigma^2(p + 1)/N$.

Next

Lab 1 (Wednesday, Rodrigo) — numpy, scikit-learn, the training/validation/test protocol, K -fold cross-validation, and the numerical example of this lecture recomputed from scratch.

Lecture 2 (Friday) — the perceptron and its mistake bound; gradient descent convergence for smooth convex losses; support vector machines and the hinge loss; logistic regression, its likelihood, and Newton's method.

Reading — Hastie, Tibshirani & Friedman, *The Elements of Statistical Learning*, §2.1–2.4, §3.1–3.2, §7.1–7.3.